Assumptions Register

ICT 5143 Capstone, Problem 8: reliable intersectional fairness audits (ACSPublicCoverage, CA 2018)

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Version 1.0 · evidence from the executed notebook of 25 October 2026

Every assumption the analysis relies on is listed with where it is used, how it was checked and the evidence from this run. "Where" refers to notebook sections. Status: checked verified by a test in the notebook; partly checked some evidence, not conclusive; violated / limitation evidence shows it fails or limits the conclusions; untested no check was run; value judgement a normative choice that needs stakeholder justification; by design true by construction of the code. Rows were reviewed against manuscript v1.0 and the literature review; the assumptions the literature raises (exchangeable groups, A2; no causal reading of the graph, P1 and E1) are included.

Key assumptions (the 12 that affect the conclusions)

The register has 39 rows. 15 of them are routine: they are verified (7) or true by construction (8). Another 12 are partly checked. The 12 below are the ones a reader should weigh: 4 are known to fail, 3 are value judgements and 5 are untested. Full details follow in the complete register.

ID	Assumption and what the evidence shows	Status
F4	Validation-fitted thresholds transfer to test: **all 116 (CA) and 27 (TX) validation-time pooled tolerance claims failed on test**	violated / limitation
A3	Attribute-structured pooling captures the deficit: it masks interaction-only deficits (FTS 0.23–0.37)	violated / limitation
A4	Empirical-Bayes plug-in intervals are adequate: struct covers 0.665 at nominal 0.90 (oracle 0.905)	violated / limitation
C1	The synthetic calibration design is representative: ACC falls from 0.830 to 0.32 on the real design (addressed by design-matched calibration, C5)	violated / limitation
F1	Equality of opportunity is the right primary criterion for outreach	value judgement
F2	$\epsilon = 0.10$, 30% capacity and a 0.02 utility budget are acceptable to stakeholders	value judgement
C2	Youden's J (equal weight on false certification and false alarm) selects the procedure	value judgement
D2	PUBCOV is reported without group-dependent error	untested
S3	The replicate-weight variance formula applies to the test subsample	untested
P2	The BN discretisation loses little information	untested
A1	Individuals within an intersection are independent (household clustering ignored in the audit likelihood)	untested
A7	The 60×60 grid represents the full-Bayes hyperposterior	untested

Complete register

ID	Assumption	Where	Check and evidence (this run)	Consequence if violated	Status
D. Data and measurement					
D1	The Folktables `ACSPublicCoverage` definition (AGEP <65, PINCP $\leq 30{,}000$, 19 features) defines the task population.	§1	Features identical cell by cell to `df_to_numpy`; 138,554 rows in both.	Results would not be comparable with the benchmark literature.	checked
D2	$Y = 1 - \mathbb{1}[\text{PUBCOV} = 1]$ records *absence of public coverage* as reported in the ACS, without systematic group-dependent misreporting.	§1	The exact reversal is verified. Reporting accuracy of PUBCOV is not modelled.	Group-dependent label noise biases TPR gaps (Fogliato et al.). The label does not measure uninsurance or eligibility.	untested
D3	Recorded RAC1P, SEX and DIS categories are used as recorded, never collapsed or inferred.	§1, §5	Nine race codes are kept: 36 intersections, 14 of 35 non-empty test cells have <30 positives.	Collapsing would hide groups. Keeping codes produces very sparse cells.	by design
D4	PUMS "not applicable" (NaN) is a meaningful category coded 0.	§1	Matches the Folktables output exactly.	A different coding would change features for minors and non-workers.	checked
D5	The 2018 1-Year PUMS vintage fetched by Folktables is the intended source.	§1	Downloaded through `ACSDataSource`; an offline zip fallback is documented.	Another vintage would change counts.	checked
S. Sampling, splitting and estimand					
S1	The household (SERIALNO) is the independent sampling unit.	§2, §11	Household-disjoint 60/20/20 splits; the bootstrap resamples households. PUMA geography and survey strata are not modelled.	Uncertainty intervals are too narrow if geographic clustering matters.	partly checked
S2	The primary estimand is the unweighted benchmark sample.	§1, §13	Survey-weighted TPRs with 80 replicate weights for the 4 sex×disability groups agree qualitatively (disabled 0.011, 0.031; non-disabled 0.280, 0.448).	Population statements for intersections would need weighting; only four groups were weighted.	partly checked
S3	Successive-difference replicate variance $\frac{4}{80}\sum_r(\hat{\theta}_r - \hat{\theta})^2$ applies, and ratio estimates need no rescaling on the test subsample.	§13	Taken from ACS documentation; not compared with published standard errors.	The weighted SEs would be wrong.	untested
S4	The test split is not used for fitting, threshold, λ or procedure selection.	§2, §9, §10	By construction: bins, CPTs, thresholds and λ use train/validation, and §9 uses only synthetic data. The analysts did see earlier versions' test results (not a blind analysis).	Optimistic test results.	by design
S5	Five refits on household splits measure *training* variability; the bootstrap measures *evaluation* variability.	§10, §11	Reported separately. Seeds are not independent population samples.	Conflating them overstates certainty.	by design
P. Predictive Bayesian network					

ID	Assumption	Where	Check and evidence (this run)	Consequence if violated	Status
P1	An augmented naive-Bayes structure is adequate for ranking. It is associational, not causal.	§3, §4.1	AUROC 0.729 ± 0.004 (LR 0.716, GBT 0.768). The structure was specified, not learned.	Weaker ranking changes who is selected, but not the audit methodology.	partly checked
P2	Fixed bins for AGEP and SCHL and training quartiles for PINCP lose little information.	§2	Not tuned; no sensitivity analysis.	Coarse bins may degrade group-specific ranking.	untested
P3	Dirichlet $\alpha = 1$ smoothing; positivity (all CPT entries > 0).	§3, §3.1	Normalisation asserted; positivity holds by construction. α is not tuned.	Zero CPT cells would make evidence combinations impossible.	checked
P4	Exact inference is implemented correctly.	§3.1	VE (two orders) = pgmpy VE = clique-tree BP = enumeration to 3.3×10^{-16}; the joint CI SEX $\perp$ {AGEP, DIS} $\mid Y$ holds.	Wrong posteriors.	checked
P5	Predicted probabilities are adequately calibrated within groups.	§4.1	Max group ECE 0.110 for BN vs 0.069 (LR) and 0.063 (GBT).	Matters for probability use; the rule uses only rankings.	partly checked
F. Fairness criterion and decision rule					
F1	Equality of opportunity (equal TPR among $Y = 1$) is the right primary criterion for a benefit-type outreach decision.	Prereg.	Justified in the problem statement; demographic parity is reported as a secondary audit.	A different criterion could rank rules differently.	value judgement
F2	Capacity 30%, tolerance $\epsilon = 0.10$ (robustness 0.05, 0.15) and a 0.02 recall budget are reasonable.	Prereg., §9.1, §12	Chosen by the analysts, not by stakeholders. Real-data conclusions are unchanged for $\epsilon \in \{0.05, 0.10, 0.15\}$.	Different stakeholders may accept different trade-offs.	value judgement
F3	Empirical ΔEOpp uses groups with ≥ 30 positives; model-based audits use every group with ≥ 1 positive.	§4, §6	Both are reported; exploratory cells are flagged.	The two gaps measure different populations of groups.	by design
F4	Group thresholds fitted on validation transfer to test.	§10	**All 116 validation-feasible selections had test ΔEOpp of 0.227–0.770 (none ≤ 0.10); out of sample on Texas, 27 of 27 (0.110–0.888).** The raw validation gap at $\lambda = 1$ is 0.48.	Validation-time fairness claims are false on new data.	violated / limitation
F6	Half of the validation households suffice both to choose λ and to certify the rule (held-out certification).	§15.4	Recall unchanged (CA 0.379 vs 0.380; TX 0.350 vs 0.350); no false claims, against 10 of 10 with same-data certification. Whether it would certify a truly fair rule is untested on real data (no rule is fair).	Lost utility, or a procedure that never certifies.	partly checked
F5	Capacity matched exactly on validation implies matched capacity on test.	§10	Test selection rates 0.292–0.305.	Small utility comparisons are confounded.	partly checked
A. Hierarchical audit models					
A1	Given the frozen rule, $k_g \sim \text{Bin}(n_g, \theta_g)$ with independent individuals within g.	§6	Household clustering is ignored by the audit likelihoods (the bootstrap respects it). No overdispersion check.	Overdispersion makes posterior intervals too narrow.	untested

ID	Assumption	Where	Check and evidence (this run)	Consequence if violated	Status
A2	Group TPRs are exchangeable (hier, full Bayes, contamination prior).	§6, §7	Real TPRs are bimodal by disability (DIS=1 ≤ 0.08). PPC $p = 0.64$–0.85 does not reject, but the range statistic is weak.	Shrinkage toward a grand mean that describes no group.	partly checked
A3	struct: additive race, sex and disability effects on the logit scale plus $N(0, \tau^2)$ interactions.	§6, §6.1	On real counts $\hat{\tau} = 0.01$ (lower bound), so the fit is effectively main effects only. In §9.2 it masks interaction-only deficits (FTS 0.23–0.37).	An interaction-type disadvantage is shrunk away.	violated / limitation
A4	Plugging in empirical-Bayes hyperparameters (hier, struct, contamination) is adequate for intervals.	§6.1, §8	struct under its own model: coverage 0.665 ± 0.014 at nominal 0.90 (oracle with true β, τ: 0.905); $\hat{\tau}$ averages 0.18 vs a true 0.30. hier gap coverage 0.000 in §8.	Over-confident intervals and false certainty.	violated / limitation
A5	Pooling bounds ($\kappa \in [0.5, 10^4]$, $\tau \in [0.01, 10]$, $\lvert\beta\rvert \leq 20$) do not drive conclusions.	§6	The τ lower bound is active for struct on real data and in 5% of the §6.1 fits. One fix (shared κ bound for fit, weights and sampling) was verified earlier.	Results at the boundary depend on an arbitrary limit.	partly checked
A6	The contamination slab Beta(1,1) and $\pi \leq \pi_{\max}$ are reasonable.	§9.1, §12	Jeffreys slab Beta($\frac{1}{2}, \frac{1}{2}$) behaves alike (mean FTS 0.444 vs 0.427, ACC 0.836 vs 0.819). $\pi_{\max}$ was swept 0.8–0.99.	The decision frontier would shift.	checked
A7	The full-Bayes hyperprior ($\mu \sim U(0,1)$, $p(\kappa) \propto (1+\kappa)^{-2}$) is adequately represented on a 60×60 grid.	§6	Grid resolution not varied.	Posterior approximation error.	untested
C. Simulation and calibration					
C1	The synthetic calibration design (16 groups, 4000 to 8 positives, base TPR 0.65, $\delta = 0.20$) is representative of the deployment design.	§8, §9, §9.2	**Out-of-sample check: the calibrated procedure's ACC falls from 0.830 to 0.32 on the real 32-group design** (FTS 0.503 vs 0.133).	The calibrated operating point does not hold where it is used.	violated / limitation
C2	Youden $J = \text{ACC} - \text{FTS}$, weighting false certification and false alarm equally, is the right selection rule.	§9	Preregistered. The winner (robust@0.8 plug-in, $J = 0.327$) and runner-up (upper 0.5, $J = 0.323$) differ by less than the Monte Carlo error.	Another weighting (e.g. penalising FTS more) selects a different procedure.	value judgement
C3	The semi-synthetic worlds (homogeneous 0.38 ± 0.015; deficit 0.20 at `2|2|1` as a cell, a pure interaction or a disability main effect) are informative scenarios.	§9.2	Labelled SEMI-SYNTHETIC. The interaction world equalises every main-effect margin exactly (mean shift 0.016).	They are scenarios, not facts about California.	by design
C5	Design-matched calibration worlds (homogeneous TPRs plus a 0.20 deficit at a random 10–60-positive group, built from the deployment population's validation sizes) represent the deficits an auditor must catch.	§15	Out of sample on TX, with deficit structures not used in calibration (interaction-only, disability main effect, thinning): $J = 0.664$ vs ≤ 0.385 for established audits. Same generator family for calibration and evaluation.	The calibrated operating point fails for other kinds of deficit.	partly checked

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C4	$R = 300$ replicates per cell are enough.	§8, §9.2	MC s.e. of a rate ≤ 0.029.	Near-ties (C2) are not resolved.	partly checked
I. Statistical inference					
I1	The household Poisson bootstrap approximates evaluation uncertainty for the fixed seed-0 model.	§11	$B = 1000$; standard but not validated by simulation.	Wrong CI widths.	partly checked
I2	The confirmatory family is the calibrated procedure vs hier plug-in and vs raw plug-in on test ΔEOpp (Holm); everything else is exploratory.	Prereg., §11	The arm follows §9's choice (plug-in rule); this clarification was recorded before §10–§11 were run. Holm $p = 1.0$ and 0.736.	Selective reporting if the family were changed after seeing results.	by design
I3	Bayesian credible intervals are not assumed to be frequentist intervals.	§6.1, §8, §9.2	Coverage is estimated by repeated simulation.	–	checked
E. Interpretation					
E1	No causal reading of BN edges or group differences.	§3	No interventions or counterfactuals are computed.	Unsupported causal claims.	by design
E2	The conclusions generalise beyond CA 2018 only qualitatively.	§14	Texas 2018: intersectional ΔEOpp 0.888 vs 0.094 by sex (seed 0), and the H1 test over 5 splits replicates the selection failure (27 of 27 pooled claims fail). Only one replication state and one rule family.	Other populations may differ.	partly checked
E3	$D = 1$ is hypothetical outreach; no real person is contacted.	all	By design.	–	by design